

Avery Thomas

Embedded Systems & IoT Firmware Engineer • 10 Years Experience • MERIQ Index: 98% (Mastered)

■ avery.thomas384@example.com | ■ +1 (214) 644-3830 | ■ Boston, MA

TECHNICAL PROFICIENCIES & CERTIFICATIONS

Core Technologies: Python, Embedded Linux, FreeRTOS, I2C/SPI/UART, PCB Debugging, C/C++, ARM Cortex

Verified Certification: Embedded Linux Developer Certificate

PROFESSIONAL WORK EXPERIENCE

Senior Embedded Systems & IoT Firmware Engineer — *Texas Instruments* (2023 - Present (3 yrs))

- Spearheaded architectural redesign of core Python service, reducing latency by 60%.
- Led a squad of 5 engineers delivering scalable microservices processing 19M requests/day.
- Implemented automated testing & CI/CD pipelines increasing deployment frequency by 42%.

Associate Embedded Systems & IoT Firmware Engineer — *Tesla* (2016 - 2023)

- Designed and maintained Embedded Linux data endpoints with 98% SLA uptime.
- Optimized database indexing and caching layers using FreeRTOS saving \$41k/yr cloud costs.

NOTABLE TECHNICAL PROJECTS

Automotive CAN-Bus Telemetry Gateway with Embedded Linux and Secure Boot [Python, Embedded Linux, FreeRTOS, I2C/SPI/UART]

Engineered end-to-end production solution leveraging Python and Embedded Linux with full test automation.

Smart Grid Power Monitoring Device with MQTT & Hardware Cryptographic Accelerator [FreeRTOS, I2C/SPI/UART, PCB Debugging, C/C++]

Architected high-resilience system integrating FreeRTOS with 99.9% uptime.

EDUCATION & CREDENTIALS

M.S. in Software Engineering

IIT Madras — 2012 - 2016 (Score: 3.46 / 4.0)